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Advanced Computer Architecture

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The student has to choose 2 out of the 3 exercises.

If all the 3 exercises will be completed, just the 2 with the lowest score will be considered.

This is an open book exam, which means the students can you use all their notes (we do not care of the format) but they cannot share their exams with anyone during the test.

At the end of the exam, when the professor will call it, each student has to upload 1 PDF file (named using the “codice persona” – CodicePersona.pdf) using the Microsoft Form share via the Zoom chat at the beginning of the exam.

In the first page of the uploaded PDF, the professor has to find the name and the surname of the student.

Problem 1 (50%)	
Problem 2 (50%)	
Problem 3 (50%)	
Total (100%)	

Problem 1

Question1 - To implement cache coherency one of the protocols is the MESI concurrency protocol implemented for a write-invalidate write-back cache. Describe briefly the algorithm (also in term of graph) and complete the following table.

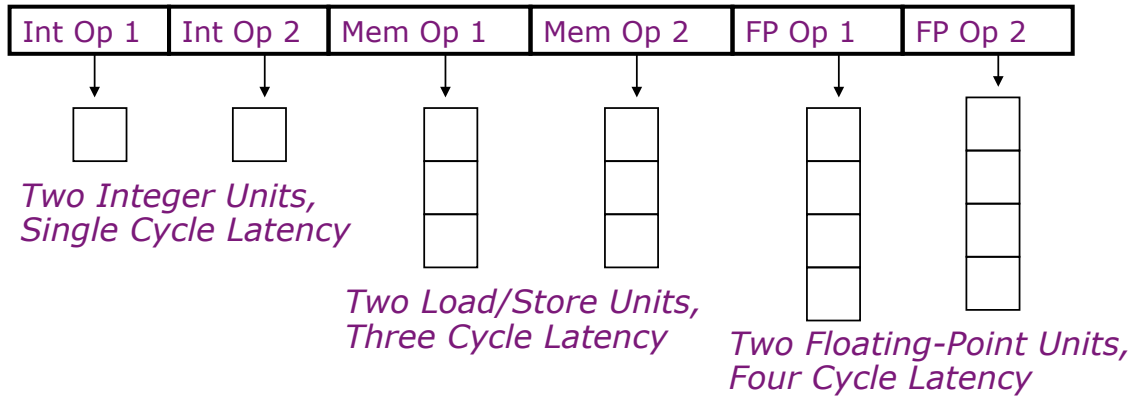
State	Cache up to date?	Memory up to date?	Others have a copy?	Cache can respond other's reads?
Modified	Yes	No	No	Yes
Exclusive	Yes	Yes	No	Yes
Shared	Yes	Yes	yes	No
Invalid	No	Maybe	Maybe	No

Question 2 - Consider the following access pattern on a two-processor system with a direct-mapped, write-back cache with one cache block and a two cache block memory. Assume the MESI protocol is used, with write-back caches, and invalidation of other caches on write (instead of updating the value in the other caches). **NOTE: you can recreate the table in excel, you can draw it on paper, you can create a "ASCII" version of it... the only thing is that, at the end, the information has to be clear and not ambiguous.**

Time	After Operation	P1 cache block state	P2 cache block state	Memory at block 0 up to date?	Memory at block 1 up to date?
0	P1: read block 1	Exclusive (1)	Invalid (0)	Yes	Yes
1	P2: read block 0	Exclusive (1)	Exclusive (0)	Yes	Yes
2	P1: write block 0	Modified (0)	Invalid (0)	No	Yes
3	P2: read block 1	Modified (0)	Exclusive (1)	No	Yes
4	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
5	P2: write block 1	Invalid (1)	Modified (1)	Yes	No
6	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
7	P2: write block 1	Invalid (1)	Modified (1)	Yes	No
8	P1: read block 0	Exclusive (0)	Modified (1)	Yes	No
9	P2: write block 1	Exclusive (0)	Modified (1)	Yes	No
10	P2: write block 1	Exclusive (0)	Modified (1)	Yes	No
11	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
12	P2: write block 1	Invalid (1)	Modified (1)	Yes	No
13	P1: read block 1	Shared (1)	Shared (1)	Yes	Yes
14	P2: read block 1	Shared (1)	Shared (1)	Yes	Yes

Problem 2

Considering the following VLIW "architecture":



Considering the following portion of assembly:

```

Loop:    LD F0, 0(R1)
         LD F6, 8(R1)
         LD F10, 16(R1)
         LD F14, 24(R1)
         FADD F4, F0, F2
         FADD F8, F6, F0
         FADD F12, F10, F14
         FADD F16, F14, F2
         SD F4, 0(R1)
         SD F8, 8(R1)
         ADD R1, 32
         SD F12, -16(R1)
         BNE R1 R2 Loop
         SD F16, -8(R1)
  
```

Question1 - Describe the corresponding VLIW code:

NOTE: you can recreate the table in excel, you can draw it on paper, you can create a "ASCII" version of it... the only thing is that, at the end, the information has to be clear and not ambiguous.

Answer:

Clock	Int1	Int2	Mem1	Mem2	FP+ (+ = ADD)	FPx (x = MUL)
1			LD F0, 0(R1)			
2			LD F6, 8(R1)			
3			LD F10, 16(R1)			
4			LD F14, 24(R1)		FADD F4, F0, F2	
5					FADD F8, F6, F0	
6						
7					FADD F12, F10, F14	
8				SD F4, 0(R1)	F16, F14, F2	
9	ADD R1, 32			SD F8, 8(R1)		
10	BNE R1 R2 Loop					
11				SD F12, -16(R1)		
12				SD F16, -8(R1)		

Question2 – How many FLOPS/cycle?

Answer: 4/12 = 1/3

Ww can also accept:

Clock	Int1	Int2	Mem1	Mem2	FP+	FPx
1			LD F0, 0(R1)	LD F10, 16(R1)		
2			LD F6, 8(R1)	LD F14, 24(R1)		
3						
4					FADD F4, F0, F2	
5					FADD F8, F6, F0	
6					FADD F12, F10, F14	
7					F16, F14, F2	
8				SD F4, 0(R1)		
9	ADD R1, 32			SD F8, 8(R1)		
10	BNE R1 R2 Loop			SD F12, -16(R1)		
11				SD F16, -8(R1)		

With a 4/11 FLOPS/cycle

Problem 3

Given the following assembly code and knowing that R0 is set to 100 and R1 is set to 0:

```
LOOP:      LD      F1  0    R0
           ADDD    F2  F1   F1
           ADDI    R1  R1   100
LOOP2:     MULTD   F2  F2   F1
           SUBI    R1  R1   1
           BNEZ    R1  LOOP2
           SUBI    R0  R0   1
           BNEZ    R0  LOOP
```

Question 1 - Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the previous code. The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer:

We are not reporting in this solution the State Machine, even if they were supposed to be included, because you can easily find them on the slides.

Said that, we do have to remember that the BHT has to be initialize and we are considering two "non-conflicting" addresses.

With the 1-BHT, we can have just two states, therefore two initialisations: Taken or Not Taken. With the 2-BHT, we can have four states/initialisations; Strong Taken, Strong Not Take, Weak Taken, Weak Not Taken

Considering the code, the loops have to be iterated, on one hand the best initialisations are: Taken for the 1-BHT and Strong Taken (even weak is ok) for the 2-BHT. On the other hand the worst initialisations are: Not Taken for the 1-BHT, and Strong Not Taken for the 2-BHT.

Listing them as possible scenarios we do have:

- a. Taken for the 1-BHT
- b. Strong Taken for the 2-BHT
- c. Not Taken for the 1-BHT
- d. Strong Not Taken for the 2-BHT

(A) will lead to:

- . First execution of LOOP2: 1 miss
- . Following executions of LOOP2 (99): 2 miss per execution - - $2 * 99 = 198$
- . For LOOP will have just 1 miss
- . Overall miss: 200

(B) will lead to:

- . First execution of LOOP2: 1 miss
- . Following executions of LOOP2 (99): 1 miss per execution - - $1 * 99 = 99$
- . For LOOP will have just 1 miss
- . Overall miss: 101

(C) will lead to:

- . All the executions of LOOP2 (100): 2 miss per execution - - $2 * 100 = 200$
- . For LOOP will have just 2 miss
- . Overall miss: 202

(D) will lead to:

- . First execution of LOOP2: 3 miss
- . Following executions of LOOP2 (99): 1 miss per execution - - $1 * 99 = 102$
- . For LOOP will have just 3 miss
- . Overall miss: 108

D is better than A which means that the worst case for the 2-BHT is better than the best solution for the 1-BHT.

Answer:

Question 2 – Is it true that the 1-BHT is always outperformed by the 2-BHT. Please, effectively support your answer.

Answer: No, the 1-BHT is not always outperformed by the 2-BHT, as an example you can think of a non-nested loops situation